What a low-cost smartphone protocol can and cannot establish about urban noise

Hanoi, 363 field measurements, three negative results
and the instrument we ended up building

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Context and research questions

The gap this project sits in

- Hanoi: dense, fast-growing, **motorcycle-dominated** traffic
- **No national strategic noise-mapping programme** and no statutory noise action plan in Vietnam
  Nguyen et al. 2025, *City Environ. Interactions*
- **One** instrumented campaign ever published for Hanoi — and it measured in **2005–2007**
  Phan et al. 2010, *Applied Acoustics* 71(5)
- Class 1 sound level meters and commercial propagation software: out of reach for most local research budgets

The question

How far does a protocol built from **three consumer smartphones** and **open data** actually get you?

The honest answer turned out to be: further than nothing, and much less far than a map.

The research question, and what is out of scope

The question, as we posed it in June — unchanged

Can urban noise be attributed to **the form of the city**, and to **the type of vehicle** passing through it? Models validated on car-centric European cities are unproven on motorcycle-heavy soundscapes.

How it was operationalised

1. Can OpenStreetMap morphology predict noise at an **unmeasured location**?
2. Does a model **transfer** between cities?
3. Does vehicle **flow from video** explain the levels?
4. What can the protocol claim with **no absolute reference**?

Out of scope, declared up front — night-time (10 points after 21:00), compliance against any standard, any prediction beyond the three sites.

And the answer, on both halves

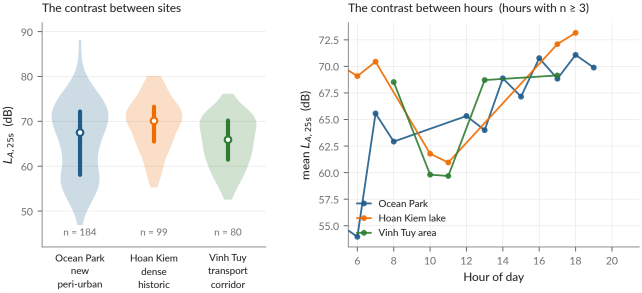
Form: no — morphology adds nothing the strict protocols can see.

Vehicle type: no — and the chain that was meant to measure it does not work.

What survives is $\log(d_{\mathrm{road}})$: distance to the road, and little else.

Data and protocol

The campaign



`data/processed/measurements.csv`

Protocol — 3 handsets, *Decibel X* (A-weighted, SLOW), 20–30 s per point, a clip $\geq$ 10 s, a 10 m GPS gate, ODK Collect $\rightarrow$ Kobo.

Published, CC-BY-4.0 — both CSVs and both XLSForms. Videos and raw Kobo exports are **not**: faces, plates, collector identities.

Metrology — the constraint everything else is built around

Three phones, cross-calibrated against *each other* and against nothing

A bias common to all three is **invisible in the data by construction**.

What that forbids

- Stating an absolute level as a fact
- Any compliance claim under QCVN 26:2010 — it requires a class 1 or 2 instrument

What it still allows

- Contrasts *between places* and *between hours* — a common bias cancels in a difference
- A *bounded* bias interval from instrumented literature: $+3.0$ **to** $+7.3\,\mathrm{dB}$

The measured quantity, and the only one:

$$L_{A,25\mathrm{s}} \;=\; 10\log_{10}\left(\frac{1}{T}\int_0^T \frac{p_A^2(t)}{p_0^2}\,\mathrm{d}t\right), \qquad T \approx 25\,\mathrm{s},\; p_0 = 20\,\mu\mathrm{Pa}$$

Not $L_{\mathrm{Aeq,\,8h}}$, **not** L_{den}, **not** L_{night} — those integrate over hours or a year.

`docs/metrology.md` · `results/tables/literature_anchoring.md` the interval is estimated, never applied

Method and evaluation protocol

The evaluation protocol — and why it is the whole story

Every model is scored under **three splits**, on exactly the same folds:

1. **Block-CV, 600 m** — square blocks in UTM 48N, `GroupKFold` on the block. The permissive one.
2. **Buffered leave-one-out, 300 m** — for each point, train on *all* points more than 300 m away. **The reference protocol.**
3. **Leave-one-site-out** — generalisation to an unseen urban typology.

Every score carries a 95 % interval bootstrapped **by block**, not by point: resampling correlated points gives an interval that is far too narrow.

Why 300 m and not less

The morphology features aggregate over a disc of **radius 300 m**. Two points 110 m apart still share **77 %** of that disc — and two either side of a cell boundary share **almost all** of it.

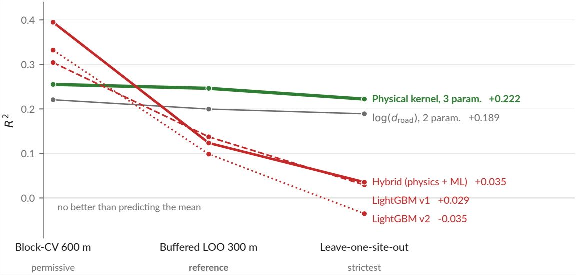
Group on 110 m cells and the model sees near-twins of its test points in training. The score is then not out-of-sample.

Roberts et al. 2017, *Ecography*

`scripts/04_evaluate_models.py`

Results

The ranking inverts — and that is the result



`models/metrics.json -- the same folds for every model`

Read across, not down

The elaborate architecture **wins the permissive split and loses both strict ones.** Three parameters is the only flat line.

What we did about it

The hybrid is **tested and rejected at this sample size** — not deferred to future work.

The delivered model

A **line**-source attenuation kernel — energy falling as $1/d$, not $1/d^2$: a street is a line of sources, not a point.

$$E = \frac{A_{\mathrm{hw}}}{\max(d_{\mathrm{hw}}, D_0)} + \frac{A_{\mathrm{res}}}{\max(d_{\mathrm{res}}, D_0)} + B$$

$$L = 10 \log_{10} E$$

$A_{\mathrm{hw}} = 4.774 \times 10^7$ $A_{\mathrm{res}} = 3.800 \times 10^7$ $D_0 = 5\,\mathrm{m}$

- $-3\,\mathrm{dB}$ per doubling of distance, not -6
- Positivity-constrained: $A_{\mathrm{hw}}, A_{\mathrm{res}}, B > 0$
- *B* **fits to** $-99.6\,\mathrm{dB}$, a boundary solution. **The background term is not identifiable here.**

`models/hybrid_physical.json` · `models/metrics.json`

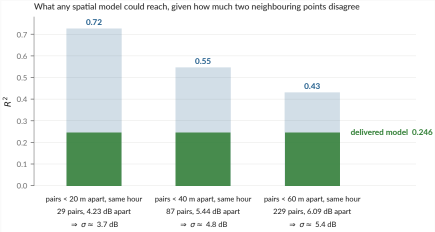
Performance under the reference protocol

R^2	**0.246**
95 % CI	[0.097, 0.339]
RMSE	6.20 dB
MAE	5.01 dB
data spread (σ)	7.14 dB

Say the error out loud

A typical prediction is wrong by **5 dB** — a factor of three in energy. Enough for a contrast, nowhere near an assessment.

“$R^2 = 0.25$ — isn’t that very low?”



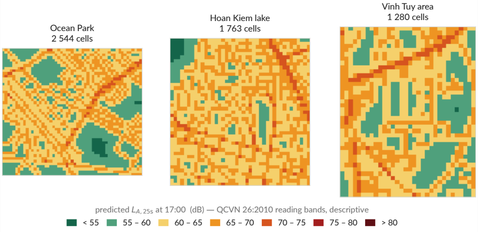
`data/processed/measurements.csv` · `docs/metrology.md`

Two points a few metres apart *at the same hour* must be predicted identically. Whatever they disagree by is variance **no spatial model can explain**: $\Delta L \sim \mathcal{N}(0, 2\sigma^2) \Rightarrow \sigma = \frac{\sqrt{\pi}}{2}\mathbb{E}|\Delta L|$, and $R^2_{\max} = 1 - \sigma^2/s^2$.

At the 40 m scale the map works at

The ceiling is near $R^2 \approx 0.54$. The delivered **0.246** is about **half** of what is attainable, not a tenth.

The map — and the caveat that goes with it



`results/maps/hanoi_noise_map.csv -- 5 587 cells`

The grid step is not the accuracy

40 m cells from a model whose typical error is **5 dB**. Bands are QCVN 26:2010, **descriptive**.

And no hour, either

17 hourly columns, **all 17 identical**: the kernel has no hour term. **Nothing varies that the model does not model.**

The three negative results

1 — A three-parameter physical law beats every learned model we built

Six-variable gradient boosting, engineered features, a hybrid architecture: all lose the reference protocol to two distances and a constant. At $n = 363$ over three sites, there is not enough signal to justify capacity.

2 — Cross-city transfer fails, and in three distinct ways

Pretrained on Sunbird Uganda, scored on our 363 points: $R^2 = -15.9$ (v1), -8.2 (convention-invariant v2). **Not a calibration offset** — removing the 26 dB mean difference still leaves R^2 negative. **v1 transfers anti-information**, $r = -0.38$: it ranks quiet and loud the wrong way round. **v2 leaves nothing**, $r = +0.08$ — not wrong, empty. Against which $\log(d_{\mathrm{road}})$ alone reaches **0.240** on the same points.

3 — Vehicle flow does not explain the levels

Non-negative energy regression on 147 matched videos returns **zero coefficients** for motorcycles and cars. Speed and source–receiver distance are not observable from a non-georeferenced camera. **Section 4 revisits how much of this is physics and how much is instrument.**

`make transfer` · `scripts/experiments/uganda_transfer.py` · `docs/negative-results.md`

Auditing our own data

We found one of our own published results was wrong

Retracted, August 2026

A LightGBM model scored $R^2 = 0.45$ under what was then described as honest spatial cross-validation. A GAMA simulation was built on it. A project-update deck presented the figure.

The grouping was on **110 m cells** while the features aggregate over **300 m**. It leaked.

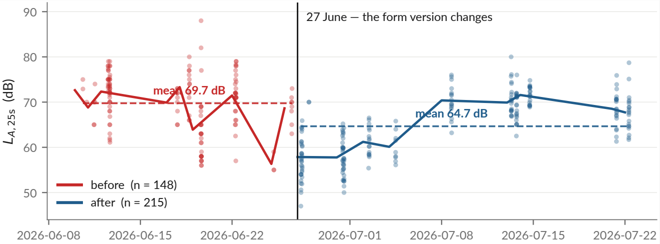
What replaced it

$R^2 = 0.246$ under buffered leave-one-out — roughly half the withdrawn figure, and the one that survives a split excluding the feature support.

- The retracted deck and map are in `docs/archive/`, **with their reason**, not deleted
- Nine corrections applied the same day
- The project pivoted from *producing a noise map* to *studying what this protocol supports*

`docs/audit/scientific-audit.md` · `docs/project-timeline.md` · `docs/archive/`

Finding 1 — a $-5.1\,\mathrm{dB}$ discontinuity on 27 June



`data/processed/measurements.csv` · `docs/audit/`

Controlling for site, hour and class: $-5.14\,\mathrm{dB}$ (SE 0.55, $p \approx 10^{-20}$). The form version changes the same day — integer readings 87 % → 20 %.

Why it matters

Before/after is predictable from $(\mathrm{lat}, \mathrm{lon})$ alone at **AUC 0.83**. An instrumental shift is then spatially structured, and a model can learn it **as morphology**. **Status: open**.

Findings 2 and 3 — the video chain, and the intervals

2 — The video chain is measuring something, but not what we think

- Detector modal split: **57 % cars, 36 % motorcycles**. In Hanoi. Reality is roughly the inverse.
- Video→measurement gaps: median 15 s, p90 56 s, **max 161 s** — for a 25 s sample
- Matched subset spans 61–80 dB against 47–88 dB overall: σ **cut by 42 %**
- Flow vs level: $r = -0.11$. A *negative* correlation between traffic and noise is not physical.

Consequence for negative result 3: it is far more likely an **instrument failure** than a finding about acoustics. It should be reported as *“the chain could not be made to work”*.

3 — The headline ranking is not statistically separated

`physical` $R^2 = 0.246$, CI $[0.097, 0.339]$ vs `dist_road` $R^2 = 0.200$, CI $[0.077, 0.266]$. The intervals overlap almost entirely. **“The physical kernel beats log-distance” is a coin flip, and the deck’s own table should show the CIs.**

What the audit leaves standing

Holds

- The evaluation protocol and the ranking inversion
- Negative result 1 — capacity does not pay at $n = 363$
- Negative result 2 — transfer fails, $R^2 < 0$
- Provenance: raw $\rightarrow$ published is lossless and checkable
- Every published number traces to one file

Needs work before submission

- The 27 June discontinuity — diagnose, then refit with the phase
- Negative result 3 — reframe as an instrument failure
- Carry the confidence intervals into every summary table
- Flag observed vs derived values in the published CSV (15 % of `construction_nearby` is inferred from the noise class)

The instrument was the problem in July, and it is still the problem in August.

Which is the argument for the next section.

From a pipeline to an instrument

Why build an application at all

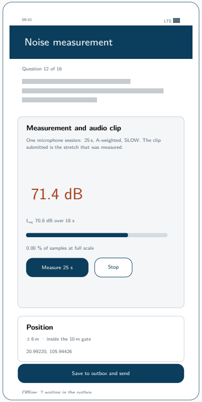
Four things the campaign could not do

1. **Control the instrument.** *Decibel X* is closed: no visible weighting, window or gain.
2. **Measure and record in one session.** Two apps meant the level of one stretch of street and the audio of another.
3. **Scale beyond three people.** Nobody pairs ODK Collect and a sound meter by hand.
4. **Record what the device was.** Handset, OS, audio source — none of it reached the dataset.

The decision, 19 August 2026

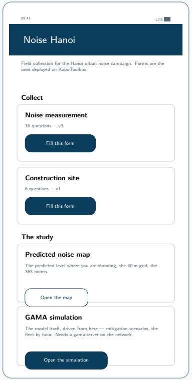
One application that **replaces the pairing** — forms, level, clip, GPS fix, submission, offline-first — and **carries the study**, negative results included.

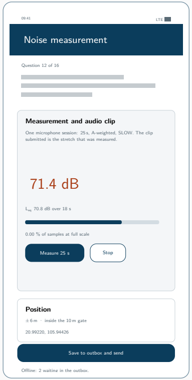
`mobile/README.md` Kotlin + Compose, one `:app` module, `minSdk` 26

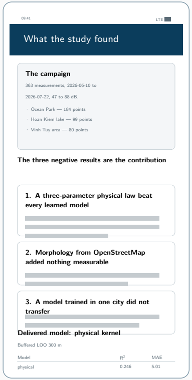


one session: level + clip

The application







Collect

both forms, offline

Measure

25 s, A-weighted

Map

the level where you stand

Results

read from `metrics.json`

Vector mock-ups drawn from the Compose source. **No device capture exists yet** — which is the last slide of this section.

What the app measures — and what it refuses to claim

The app's level is a *different quantity* from the campaign's

This handset's microphone, not a trimmed Decibel X reading. It goes in **its own field**, with the device and the audio source the platform granted. **The two never share a column.**

Verified end to end, live

- Fix, 25 s measurement, instance written, **submission accepted by `kc.kobotoolbox.org`**
- A position outside the three areas is **refused, not answered**
- 36 unit tests; weighting pinned to the IEC table

Not verified — and it is the important one

The level has never been checked against a sound source. An emulator records silence. Until it is compared to a meter, **the absolute figure means nothing** — equally true, per `docs/metrology.md`, of the campaign's own numbers.

First task on a real handset, alongside Decibel X on the three campaign phones.

Driving GAMA from the phone — how it is wired

One socket, and the phone runs nothing

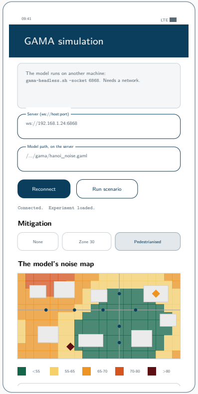
No Android port — but a **WebSocket API**: `gama-headless.sh -socket 6868`. The app is a client (`load`, `play`, `step`, `ask`, `stop`): a remote control and a screen.

Two things only driving it teaches you

A `type: gui` experiment **kills the socket** on load, and `step` without `sync` advances nothing while **every indicator reads zero**.

It redraws the model's display, it does not invent one

Every layer and colour comes from the aspect blocks of `hanoi_noise.gaml`, **in declaration order — which in GAMA is the stacking**.



ARCHITECTURE.md §5.2

Next: Clermont-Ferrand

Clermont-Ferrand — what it does and does not change

Still no reference instrument

There will be no sound level meter at Clermont either. The calibration gap does not close, and nothing said here should suggest it will: the app's level stays relative, and the half-day next to a class 2 meter remains *unbooked*, not *done*.

So what *is* the work

Maintenance and deployment, not new instrumentation. Keep the application alive, then get it collecting in a second city.

`docs/handover.md` §6, debt #2, P0-3

What the second city is actually for

- **One instrument, two cities.** The **same APK** in Hanoi and Clermont. That is the point: the instrument stops being a confound, which is the one thing the Uganda transfer could never fix.
- **Two fabrics as different as we could pick.** Two-wheelers against cars and a tram; dense organic against European planned; flat delta against volcanic relief.
- **A fourth typology, at last.** Three sites bounded everything here. A second city is the only thing that moves that number.

The comparative deployment — and why it is the decisive test

	Hanoi	Clermont-Ferrand
Fleet	two-wheelers	cars, tram
Fabric	dense, organic	European, planned
Relief	flat delta	volcanic, hilly
Monitoring	none	EU END mapping

Held constant — the **same APK**, form, accuracy gate, $L_{A,25\mathrm{s}}$, OpenStreetMap features and evaluation protocols.

Why this is stronger than Uganda → Hanoi

There, the **instrument, protocol and feature conventions all differed too**: "transfer fails" and "the datasets are not commensurable" are not separable. One app in both cities and **the instrument stops being a confound**.

And a ground truth, finally

Clermont-Ferrand falls under EU END strategic mapping — the first **official, instrumented reference map** to compare against.

The two things a reviewer will press on

1 — No absolute reference. Was half a day really out of reach?

Three handsets calibrated against *each other* and against nothing. `docs/metrology.md` owns this squarely — but an acoustics reviewer will not ask whether we *knew*; they will ask whether **one side-by-side reading against a class 2 meter** was genuinely unobtainable.

It is the right question, and the honest answer is that a **half-day rental** would settle it. **One reading turns "calibrated in relative terms" into "calibrated"**, and every level inherits it. The cheapest item anywhere in this work.

2 — Three sites are three typologies, and that binds harder than the R^2

Our own text says it: the number of **independent morphological configurations** is close to **three**, whatever the number of points. $n = 363$ does not buy $n = 363$ worth of generalisation.

The hard limit is the design, not the R^2. **A fourth and fifth contrasting fabric would do more than any modelling change.**

Conclusion

What this work established

Findings

1. At $n = 363$ over three sites, **three physical parameters beat every learned model** we built. Capacity does not pay.
2. **Cross-city transfer fails** — $R^2 < 0$, with matched features.
3. The video chain **could not be made to work**; that is an instrument result, not an acoustic one.
4. A cross-calibrated smartphone protocol supports **contrasts**, never levels.

Practices that made those findings trustworthy

- The buffer matches the feature radius — always
- Baselines and an ablation on *the same* splits
- The delivered model is chosen **by code**
- Every number has **exactly one home**
- Retracted work is archived **with its reason**
- Nothing is predicted outside the sampled envelope

The deliverable is a **protocol, an instrument and an honest error bar** — not a noise map of Hanoi.

Thank you

`github.com/Colauz/noise-modelling-hanoi`

Everything on these slides rebuilds from the repository:

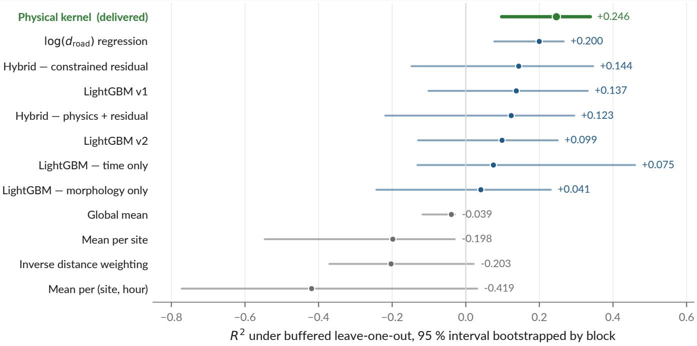
`make features` `make models` `make results` `make report`

the deck's own figures: `scripts/12_presentation_figures.py`

Backup — every formula this talk rests on

What	Definition
The measured quantity	$L_{A,25\mathrm{s}} = 10\log_{10}\left(\frac{1}{T}\int_0^T \frac{p_A^2(t)}{p_0^2}\,\mathrm{d}t\right), \quad T \approx 25\,\mathrm{s}, \quad p_0 = 20\,\mu\mathrm{Pa}$
A-weighting, IEC 61672	$A(f) = 20\log_{10}\frac{12194^2 f^4}{(f^2+20.6^2)\sqrt{(f^2+107.7^2)(f^2+737.9^2)}\,(f^2+12194^2)} + 2.00$
Levels never average	$L_\Sigma = 10\log_{10}\sum_i 10^{L_i/10}$ decibels add in energy — which is why everything below works on E
The delivered model	$E = \frac{A_{\mathrm{hw}}}{\max(d_{\mathrm{hw}}, D_0)} + \frac{A_{\mathrm{res}}}{\max(d_{\mathrm{res}}, D_0)} + B, \quad L = 10\log_{10} E, \quad A_{\mathrm{hw}}, A_{\mathrm{res}}, B > 0$
Why $1/d$, not $1/d^2$	A street is a **line** of sources, not a point: $-3\,\mathrm{dB}$ per doubling of distance, not -6
The traffic scenario	$L(k) = 10\log_{10}\left(k\,(E - E_{\mathrm{res}}) + E_{\mathrm{res}}\right)$ **not** $L + 10\log_{10} k$
Reference protocol	For each point i, train on $\{j : d(i,j) > r\}$ with $r = 300\,\mathrm{m}$ = **the feature aggregation radius**
The ceiling on R^2	$\Delta L \sim \mathcal{N}(0, 2\sigma^2) \Rightarrow \sigma = \frac{\sqrt{\pi}}{2}\,\mathbb{E}\lvert\Delta L\rvert, \quad R^2_{\max} = 1 - \sigma^2/s^2$

Backup — every model, with the interval that qualifies it



`models/metrics.json -- 2000 resamples, bootstrapped by block`

This is audit finding 3, drawn

`physical` and `dist_road` overlap almost entirely. "The physical kernel beats log-distance" is a **coin flip** at this sample size. Other protocols: `models/model_comparison.md`.

Backup — the absolute-bias interval

Anchor	Instrument	Quantity	Raw gap	Corrected
Gelb & Apparicio 2019, HCMC	dosimeters + GPS	$L_{\mathrm{Aeq,1min}}$	$+9.3$	$+7.3$
Phan et al. 2010, Hanoi	RION NL-21/22	L_{den}	$+7.0$	$+3.0$
Phan et al. 2010, Hanoi	RION NL-21/22	L_{night}	-3.0	-3.0
IOHE, 12 arteries (press)	undocumented	daytime mean	$+8.6$	$+7.6$

Conclusion

Plausible absolute bias: **$+3.0$ to $+7.3\,\mathrm{dB}$** (peer-reviewed sources, night excluded — $n = 10$ is too thin). **Corrected gap** = raw gap minus the difference the quantity alone explains: it approximates the residual instrumental bias, and **is reported and never applied**.

`results/tables/literature_anchoring.md` · `scripts/06_anchor_literature.py`

Backup — what is published, and what is not

Dataset	Size	Published	Why / licence
Field measurements	363 pts, 3 sites	yes	CC-BY-4.0
Vehicle counts from video	147 videos	yes	CC-BY-4.0
Survey forms (XLSForm)	2 forms	yes	CC-BY-4.0
Traffic videos	147, 6.0 GB	no	faces and plates
Raw Kobo exports	—	no	collector identities
OpenStreetMap extract	49 146 buildings	no	regenerable; ODbL
Sunbird Uganda 61K	61 k clips	no	gated upstream

Stated plainly in the repository

No ethics review was requested for the video collection, and none is claimed. Only non-identifying aggregates are released. The retention decision is still **open**, and is a supervisor decision.

`docs/data-sources.md` · `LICENSE, LICENSE-DATA`